

LESSON 2: COMPUTER INTERACTION

Computers interact with each other by sending and receiving information just like their internal components. Information exchange follows similar principles as those that apply inside the computer. It, therefore, follows that there must be signals, a medium through which signals will travel, a way to organize these transmissions and addresses that indicate source and destination of transmitted information.

NETWORKS

A network is a system of interconnected computers and devices that are capable of exchanging information. Networks vary depending on their size, area of coverage, and the technology used to transmit information. The smallest network may consist of two devices in one room, while the largest network is the internet a system of communication that consists of billions of devices and spans the entire globe.

The method of information transfer depends on the technology used. In some cases, transmission occurs through wired connections where data is transmitted as electrical or light signals across wires and cables. Conversely, there is wireless transmission, which uses radio waves to send signals through the air.

A NETWORK IS A SYSTEM OF INTERCONNECTED COMPUTERS AND DEVICES CAPABLE OF EXCHANGING INFORMATION.

PERSONAL AREA NETWORK

The smallest network is a personal area network, PAN. It typically consists of a personal computer connected to peripherals and other devices in its vicinity. Any personal computer can be used to establish a PAN.

Typically, the PAN is centered around an individual and the personal computer being used. Connections in a PAN can be wired, wireless or a combination of both.

A PERSONAL AREA NETWORK IS A NETWORK CENTERED AROUND AN INDIVIDUAL.

EXAMPLE 2.1

Sarah sat in her school's library. She had some homework she needed to finish and print. She was using her tablet, which was connected to a digital pen, her earbuds and the library's printer. Sarah's devices were connected in PAN through wireless connections.

These short-range wireless connections are based on Bluetooth technology. Each device has a Bluetooth antenna capable of sending and receiving radio wave signals. Most Bluetooth antennas are only capable of a range of about 10 meters, they are referred to as Class 2 Bluetooth antennas and are a good choice for short-range communications. On the other hand, Class 1 Bluetooth antennas are capable of transmissions up to a range of 100 meters.

EXAMPLE 2.2

The librarian also had her own PAN. But in her case, the PAN was wired rather than wireless. Her desktop was connected to a printer, scanner, and a smartboard by cables. This allowed her desktop to exchange information with devices in its vicinity.

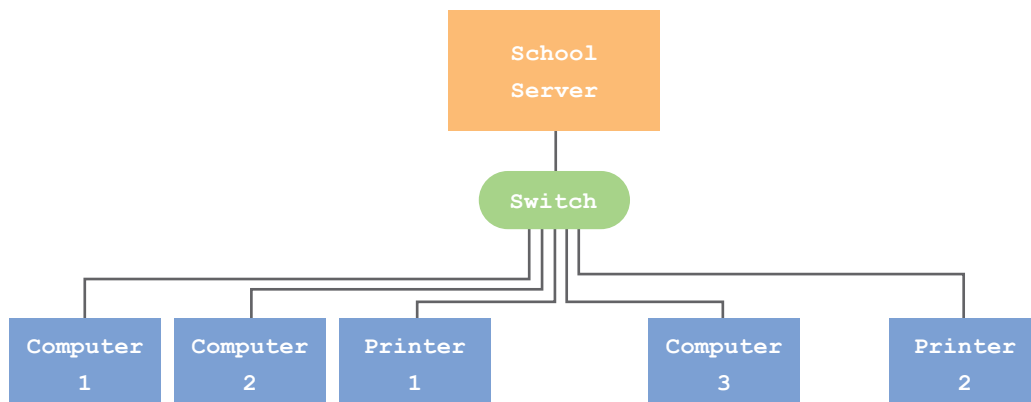
LOCAL AREA NETWORK

A local area network, or LAN, is larger than a PAN and covers a wider area such as a home, office building, or school campus. A LAN is more extensive than a PAN and serves a larger group of users.

EXAMPLE 2.3

The librarian can use her desktop to connect to other computers at school. This includes the school server which holds the school's database and information management system. By having access to the server, the librarian gains information about the inventory of books in the library, which books have been checked out and by who, as well as overdue to books. This information is shared with the school principal and teachers. Even students can login to view their list of checked out and overdue books.

Our hypothetical school LAN consists of a central server and all the computers connected to it at school. Connections are established by wires and cables, specifically Ethernet cables. This is a wired LAN, since all its connections are in the form of cables.



The figure shows the school as one local area network that consists of 3 computers, 2 printers and a server all of which connect and communicate via one switch.

Switches are networking devices that connect multiple computers and other devices within a network. They receive information from one device and forward it to the appropriate destination within a LAN.

A SWITCH IS A NETWORK DEVICE THAT CONNECTS MULTIPLE COMPUTERS AND DEVICES WITHIN A LAN.

WIDE AREA NETWORK

Wide area networks, WAN, are much larger. They vary in size of coverage from city-wide to as wide as our whole planet. The internet is the largest and most widely known example of a WAN. It is a global network of interconnected smaller networks that spans across the entire planet.

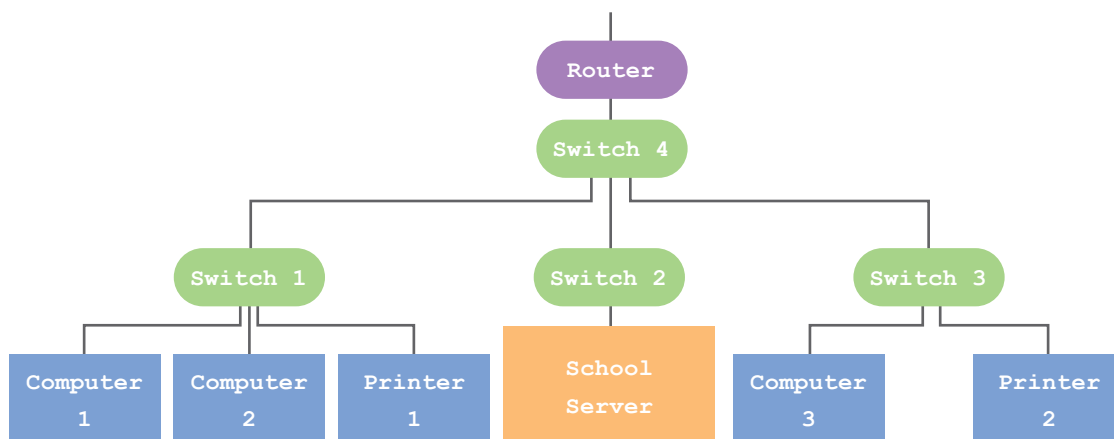
While a switch is capable of handle connections inside a LAN, it cannot perform the tasks necessary to handle connections outside its LAN. For such tasks, a router is needed. Routers act as gateways to the internet and its traffic controllers. When information has to pass outside a LAN, the router identify its destination and the best path to get there.

A ROUTER IS A NETWORK DEVICE THAT ACTS AS A GATEWAY TO THE INTERNET AND AS ITS TRAFFIC CONTROLLER.

EXAMPLE 2.4

In the previous example, we assumed a small school. A larger school would have many more computers, printers and other devices. Therefore, instead of having a large LAN, it is more appropriate to divide the school into smaller LANs.

Each section of the school would have its own switch and form its own LAN. To connect these smaller LANs, another switch would be needed forming one larger LAN. The addition of a router provides the school with access to the internet.



In this figure, the whole school campus is depicted as 1 larger LAN consisting of 3 smaller LANs. Each smaller LAN is established by a switch. All three switches are connected by Switch 4. The router provides connectivity to the internet.

The internet is considered to be the widest network. The backbone of the internet is a complex series of submarine fiber optic cables that connect all the continents and transmit information in the form of light signals. As of early 2023, there are 552 active and planned submarine fiber optic cables used for the internet. These cables span over 1.4 million kilometers and carry more than 95% of the world's international internet traffic.

THE INTERNET IS A COMPLEX SYSTEM OF NETWORKS WORKING TOGETHER TO PROVIDE COMMUNICATION ACROSS THE GLOBE.

But the internet does not solely rely on fiber optic cables for signal transmission, it also employs satellite and telephone communication networks, such as land lines and the cellphone grid.

ADAPTERS

The three common forms of energy used to as communication signals are electrical, light and radio waves. Electrical signals and light signals are used to transmit information in wires, typically copper wires for electrical signals and fiber optic cables for light, while radio signals are used to transmit information through the air. Each type of signal requires a specific type of hardware to handle its conversion but in general, communication channels are broadly categorized as wired and wireless.

THERE ARE TWO BROAD CATEGORIES OF COMMUNICATION CHANNELS WIRED AND WIRELESS.

Since the signals of communication traveling in a network are different from the electronic digital signals of the computer, signals must be converted. This is the task of the network adapter, a hardware component of the computer interface. The job of the adapter is to convert the internal signals of the computer to the external signals of the network and vice versa. As such, the adapter is considered an I/O device.

EACH ADAPTER CONVERTS INTERNAL SIGNALS TO A SPECIFIC TYPE OF EXTERNAL SIGNAL INTENDED FOR TRANSMISSION.

Wired Adapters

Each of the adapters converts electronic digital signals into a different type of transmission signal. They typically connect via a port.

1. Land line Modems convert between the electronic digital signals of the

computer and the continuous (analog) electrical signals used to transfer information through telephone land lines. They connect LANs to the internet.

2. **Ethernet network interface card, NIC**, converts between the electronic digital signals of the computer and another electronic digital signals used to transfer information through long cables. It also connects devices in LANs and the connect LANs to the internet.
3. **Fiber optic NIC** typically includes a fiber optic transceiver, which converts the electrical signals used by the device into optical signals that can be transmitted over the fiber optic cable. It is commonly used in data centers where high-speed data transmission and reliability are critical.

Wireless Adapters

The following three adapters convert between the electronic digital signals of the computer and radio waves, which are used to transfer information through the air. They capture and emit signals using an antenna. One main difference between them is their range of transmission.

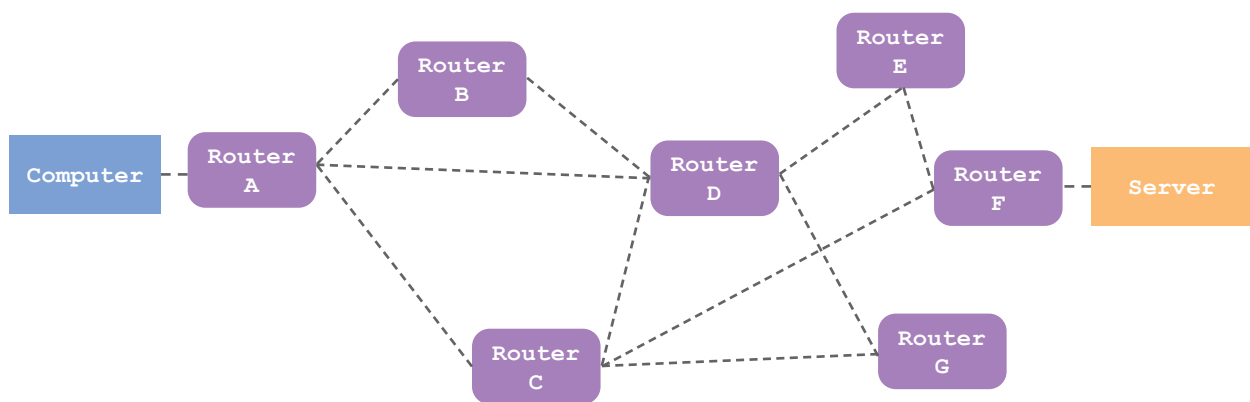
1. **Cellular Modem** is included as a component in smartphones and some tablets and is designed to transmit signals over several kilometers and is used to connect to the internet via cellular networks.
2. **WiFi NIC** can transmit signals up to 45 meters indoors and up to 150 meters outdoors. It is typically included as an internal component in most personal computers or can be externally connected to one. It is used to connect to LANs and the internet.
3. **Bluetooth Adapter** can transmit special type of radio wave signal, referred to as frequency-hopping spread spectrum (FHSS) used for short distance of upto 10 meters indoors and is used to connect to nearby devices. This allows the computer to established a personal area network between devices in close proximity such as Bluetooth-enabled earphones, headphones, speakers, television sets and other personal computers.

BEST ROUTE

The internet is a busy place. Once the signal leaves the source computer, depending on the distance it must travel, it will likely have to pass through several routers before arriving at its destination. The path the signal takes from source to destination is determined by routers.

EXAMPLE 2.4

A computer sends a request to a server. Router A is acting as the computer's gateway to the internet. Therefore, Router A needs to know the best route to send the signal to the server.



One method of determining the best route for delivering a signal to its destination is to count the number of routers the signal must pass through before arriving. This is referred to as the hop count. Every router keeps a routing database of all the possible routes and their metrics, such as the hop count.

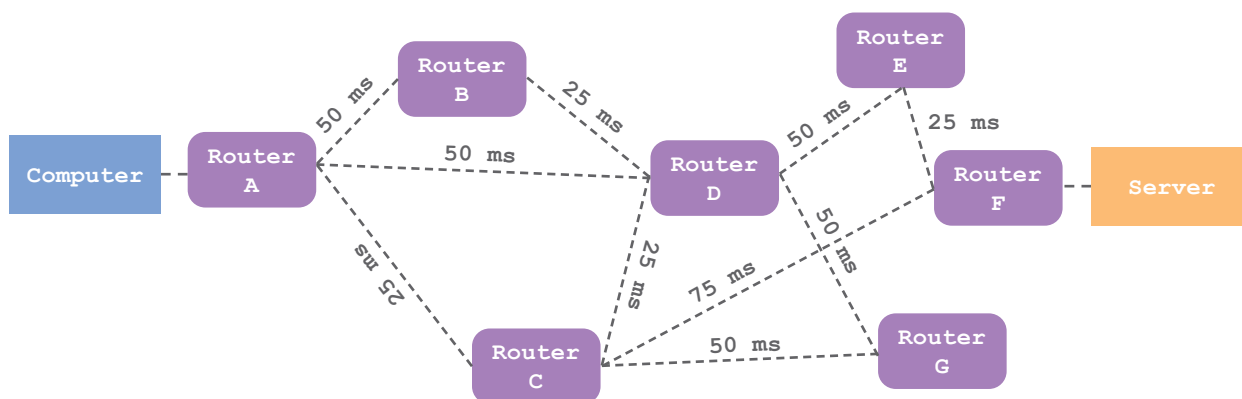
Based on this metric, Router A would take route option 3 because it has the fewest hops from source to destination.

Route Option	Hop 1	Hop 2	Hop 3	Hop 4	Hop Count
1	B	D	E	F	4
2	D	E	F		3
3	C	F			2

Another metric routers can use for determining the best paths is latency, or the time it takes for a signal to travel from one point in the network to another. One common method for measuring latency is the ‘ping’, which measures the round-trip time for a signal to travel from the source to the destination and back.

In some network configurations, routers may ping their neighboring routers. These latency measurements are then shared with other routers, which use them to choose the best route for sending signals.

A long ping indicates either that the distance between the source and destination is long or that there are delays in processing the signal, due to either heavy traffic or a slow router.



The figure shows the latency time in milliseconds between routers. Router A would require such information to determine the best route to its destination, the server. Total latency is estimated as the sum of all intermediary latencies.

Route Option	Hop 1	Hop 2	Hop 3	Hop 4	Total Latency
1	B: 50	D: 25	E: 50	F: 25	150 ms
2	D: 50	E: 50	F: 25		125 ms
3	C: 25	F: 75			100 ms

According to the database information, route option 3 remains the best route to the destination. This route not only has the fewest hops, 2, but also has the shortest latency, 100 ms.

THE BEST PATH IS DETERMINED USING METRICS SUCH AS HOP COUNT AND LATENCY.

What if Router C becomes congested with many requests to route signals through? What if the latency between Router A and Router C becomes 100 ms? In such a case, route option 3 would not seem to be so desirable having a total latency of 175 ms, the highest among all three options. Instead, route option 2 becomes the best route to the destination.

Some cases are not as clear. What if the latency between Router A and Router C becomes 50 ms? Now route options 2 and 3 have the same total latency, 125 ms.

In general, routers differ based on the decision-making program used to determine the best route. Some programs place greater importance on hop count, while others favor latency, and still others may use yet another metric.

TRUST

When transmitting signals across the internet, or between any two devices, trust must be established. The user must be able to trust that the signals are transmitted without any unintentional error and without any intentional change or theft. The transmission must maintain the integrity and security of the message it is carrying.

**THE USER TRUSTS THAT THE TRANSMISSION WILL MAINTAIN THE
INTEGRITY AND SECURITY OF THE MESSAGE.**

ERRORS

Errors in data during transmission can happen for many reasons, which, for the most part, are generally related to the phenomena of attenuation and interference.

Loss of strength, or attenuation, can occur with any form of signal in any medium if that signal is required to travel great distances. The longer the distance, the more likely it is that the signal will lose its strength, whether it is light in a fiber optic cable, electricity in a copper wire, or radio waves traveling through air.

**ERRORS IN DATA TRANSMISSION HAPPEN GENERALLY DUE TO
ATTENUATION OR INTERFERENCE.**

Another common cause of errors is interference. There are several reasons for interference, such as storms, buildings, neighboring wires, and cosmic and solar radiation.

Storms and severe weather can interfere with and disrupt radio waves, while obstacles such as buildings can block or reflect radio waves, creating multiple paths of waves that interfere with each other.

Nearby electrical wires or electronic devices can generate electromagnetic radiation that interferes with the signals, particularly electrical signals in copper wires.

Other forms of energy, such as cosmic and solar radiation, can be problematic for satellites in low earth orbit and other devices, such as computers and routers located at high altitudes.

PROTECTION

There are methods used to protect against errors and decrease the risk of their occurrence. For signal attenuation, repeaters are placed at a certain distance along the path of propagation of the signal, whether it is light in fiber-optic cables, electricity in copper wires, or radio waves traveling through the air. Repeaters are devices that capture, amplify, and retransmit the same signal, so signals whose energy is weakening become restrengthened.

Shielding works to protect against interference in copper wires and satellites. For copper wires, adding an insulating layer around the wires helps shield the signals inside from outer, neighboring interference. As for satellites, various types of shielding and design techniques are used to protect the internal electronic circuits from cosmic, and to some extent, solar radiation.

REDUNDANCY IN DATA MEANS MAINTAINING ADDITIONAL COPIES OF THE ORIGINAL DATA.

A simple and common strategy used to protect from complete loss of information is redundancy, a term which means having more of the same. When any transmission occurs, whether it is between components of a computer or between computers, only a copy of the original information is sent. The original information, or file, stays put. It is copied, and that copy is the signal that is transmitted. If the transmitted information is lost or disrupted, the original information is still safe and can be recopied and retransmitted.

DETECTION

When a signal is transmitted, the receiving computer sends back an “All is Good” signal to the transmitting computer, which basically means that “I have received your message.” This is referred to as an acknowledgement. If the acknowledgement is not received, the sender will assume that the message was lost and will transmit another copy.

If the message does arrive, then it must be checked for errors. There are several methods of error checking, one common method is a checksum. A checksum is a value calculated from the data that is sent along with the data itself. The receiver can calculate its own checksum from the received data and compare it to the one that was sent. If they match, the data is correct; if not, an error occurred.

THE ACKNOWLEDGEMENT AND THE CHECKSUM ARE TWO METHODS OF DETECTING ERRORS IN DATA TRANSMISSION.

EXAMPLE 2.5

The checksum works at the byte level. Let's assume that the message "Hello" is going to be sent. Every character in the alphabet has an 8-bit binary code to represent it. This byte actually represents the number assigned to the character.

H	e	l	l	o
0100 1000	0110 0101	0110 1100	0110 1100	0110 1111
72	101	108	108	111
Checksum	$72 + 101 + 108 + 108 + 111 =$			
	500			

In this case, the values for each of the 5 characters in the word "Hello" are added. The sum, referred to as the checksum, is sent along with the message. The computer on the receiving end will repeat the checksum calculation on the message it received. If the receiver's checksum is equal to the sender's checksum then the signals are all correct and no errors have occurred during transmission. The message the receiver has is the same as the one sent by the sender.

ENCRYPTION

Just like we expect to maintain the integrity of a message during transmission, we also expect that the transmission will be secure. The message should arrive error-free and it should not be amenable to hacking. Hacking means gaining unauthorized access to a computer, network or transmission, with the intention of changing, stealing or destroying data.

HACKING IS GAINING UNAUTHORIZED ACCESS TO A COMPUTER, NETWORK OR TRANSMISSION, TO CHANGE, STEAL OR DESTROY INFORMATION.

One way to keep a message secure from hackers is to encrypt it. Encryption changes the message in such a way as only the sender and receiver can read it. So even if hackers gained access to the message they would not know what to do with it because they cannot read it.

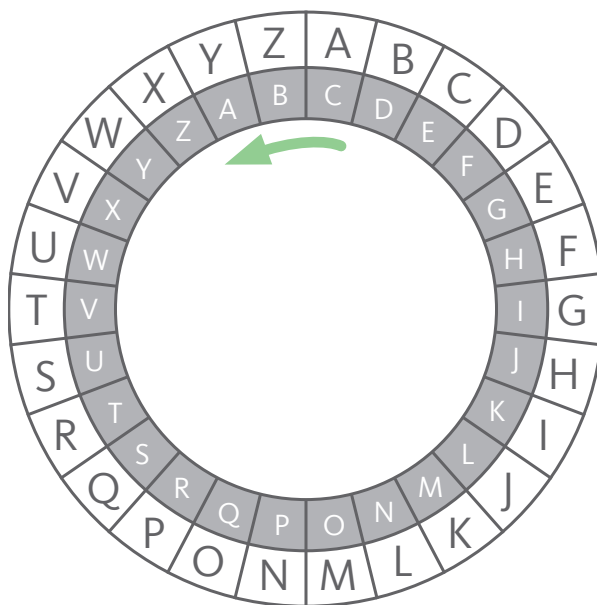
ENCRYPTION IS THE PROCESS OF REWRITING A MESSAGE IN A WAY THAT CAN ONLY BE READ BY A USER WHO HAS THE KEY.

Encryption is a process that transforms a message from a readable form, plaintext, into an unreadable form, ciphertext. The transformation is based on a set of steps, an algorithm, that determines how the message will be changed. The encryption key determines by how much the message will be changed.

EXAMPLE 2.6

The Caesar cipher is a simple encryption algorithm that dates back to the time of the Roman Empire. It is named after Julius Caesar, who is said to have used it to protect the confidentiality of his military messages. This allowed Caesar to send messages across the span of the empire with the security that should a message fall into the wrong hands it could not be read.

The Caesar cipher works by shifting each letter in the plaintext by a fixed number of positions in the alphabet. A Caesar cipher disk, is a tool used to encrypt and decrypt messages using the Caesar cipher. It consists of a circular disk or wheel with the alphabet printed on it, typically in two concentric rings. The outer ring represents the plaintext alphabet, while the inner ring represents the corresponding ciphertext alphabet after applying a specific shift value.



A Caesar cipher disk whose inner ring is turned 2 positions counterclockwise. 'A' on the outer ring corresponds to 'C' on the inner ring, 'B' corresponds to 'D', 'C' corresponds to 'E', and so on. To encrypt a message replace each letter of the plaintext on the outer ring with the corresponding letter of the ciphertext on the inner ring. In this case, the word "HELLO" would be encrypted as "JGNNQ".

The encryption algorithm would be to shift the inner wheel in a certain direction by a certain number of positions. The key would be the direction and number of positions to shift, which, in the case of this example, is 2 positions counterclockwise.

Encryption algorithms and keys are far more elaborate when considering computer-to-computer data transmission. Every communication application would have its own encryption algorithms and keys that it uses to ensure that all messages and information sent by it are secure.

Although this presents a very simplified view of computer-to-computer interaction, or data transmission, it does emphasize the importance of establishing trust by ensuring the integrity and security of transmitted information.

Career Tips

Cybersecurity is a rapidly evolving field that encompasses various aspects of protecting computer systems, networks, and data from unauthorized access, attacks, and damage.

Ethical hackers, also known as ‘white hat’ hackers are professionals who are authorized to identify vulnerabilities and weaknesses in computer systems, networks, or applications with the goal of strengthening their security measures.

SUMMARY

A network is a system of interconnected computers and devices capable of exchanging information. Networks vary depending on size, area of coverage, and the technology used to transmit information.

The smallest network is a personal area network, PAN, which is a network centered around an individual. Any personal computer can be used to set up a PAN. It can be wired, wireless or a combination of both.

A local area network, LAN, is larger than a PAN and covers a wider area and serves a larger number of users. A switch is a network device that connects multiple computers and devices within a LAN.

Wide area networks, WAN, vary in size and coverage. They consist of multiple LANs and can provide coverage for a city. The largest WAN is the internet. The internet is a complex system of networks working together to provide communication across the globe. A router is a network device that acts as a gateway to the internet and as its traffic controller.

The adapter is a computer component that converts the internal signals of the computer to the external signals of the network and vice versa. Each adapter converts internal signals to a specific type of external signal intended for transmission. There are two broad categories of communication channels wired and wireless. Wired adapters are: land line modems, ethernet NICs, and fiber optic NICs. Wireless adapters are: cellular modems, WiFi NICs, and Bluetooth adapters.

Routers are tasked with identifying the best route between source and destination. The best path is determined using metrics such as hop count and latency.

Transmission must maintain the integrity and security of the message. Errors in data transmission happen generally due to attenuation or interference. To protect against signal attenuation, repeaters are used to amplify, and retransmit signals whose energy is weakening. To protect against interference, shielding is used around copper wires and on satellites.

The acknowledgement and the checksum are two methods of detecting errors in data transmission. If the acknowledgement is not received, the sender will assume that the message was lost and will retransmit. If the receiver's checksum is not the same as that of the sender, then there is an error in the message.

Encryption is the process of rewriting a message in a way that can only be read by a user who has the key. Every encryption requires an encryption algorithm and key to transform the message from readable to unreadable forms.

Activity 2.1

Read the following statements and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>F</u>	A network does not act as a system.
<u>T</u>	The smallest network is a personal area network.
<u>T</u>	The internet is a complex system of networks.
<u>T</u>	The adapter is considered an I/O device.
<u>F</u>	Only two forms of energy are used to convey signals: light and electricity.
<u>T</u>	Network interface cards and modems are different types of adapters.
<u>F</u>	The best route is determined based only on the hop count.
<u>F</u>	Errors in data transmission happen only due to attenuation.
<u>T</u>	The checksum is one method to detect errors.
<u>F</u>	A key is not required to encrypt a message.

Activity 2.2

Complete the sentences below.

1. A network is _____.
 - a) a device that stores information
 - b) a technology used for sending signals
 - c) a system of interconnected computers and devices capable of exchanging information

2. The internet transmits information _____.
 - a) only through wired connections
 - b) only through wireless transmissions
 - c) using a combination of both wired and wireless transmissions.

3. The primary function of a switch in a network is to _____.
 - a) serve as a power supply.
 - b) receive and forward signals within a LAN
 - c) connect a LAN to the internet

4. The device used to handle connections outside a LAN is _____.
 - a) a server
 - b) a switch
 - c) a router

5. The backbone of the internet is formed by _____.
 - a) a complex series of submarine fiber optic cables
 - b) satellite and telephone communication networks
 - c) routers and switches

Activity 2.3

Answer the questions below.

1. What are the three common forms of energy used for communication signals?
 - a) Magnetic, sound and light
 - b) **Electrical, light and radio waves**
 - c) Heat, mechanical and electrical

2. What are the two broad categories of communication channels?
 - a) Internal and external
 - b) Visible and invisible
 - c) **Wired and wireless**

3. What is the primary function of a network adapter?
 - a) **To convert between internal computer signals and external network signals.**
 - b) To convert AC power to DC power.
 - c) To increase the speed of the computer processor.

4. Land line Modems convert electronic digital signals to what kind of signals?
 - a) Radio waves
 - b) **Continuous (analog) electrical signals**
 - c) Frequency-hopping spread spectrum signals

5. Bluetooth Adapters can transmit a special type of radio wave signal referred to as?
 - a) **Frequency-hopping spread spectrum (FHSS)**
 - b) Continuous wave spectrum (CWS)
 - c) Modulated wave spectrum (MWS)

Activity 2.4

Complete the sentences below.

1. The role of a router in internet connectivity is _____.
 - a) to send signals directly from the source computer to the destination server.
 - b) to store data from the source computer.
 - c) to determine the path the signal takes from source to destination.

2. Every router keeps _____.
 - a) a record of all the users in the network.
 - b) a routing database of all the possible routes and their metrics.
 - c) a list of all the signals it has processed.

3. In networking, a ping is used for _____.
 - a) measuring the round-trip time for a signal to travel from the source to the destination and back
 - b) counting the number of routers the signal has passed through
 - c) identifying faults in the network

4. In networking, latency is the _____.
 - a) time it takes for a signal to go and come back
 - b) distance between source and destination
 - c) time it takes for a signal to arrive at its destination

5. In general, routing algorithms use _____.
 - a) different metrics
 - b) only hop count as a metric
 - c) the same metrics

Activity 2.5

Answer the questions below.

1. What can cause signal attenuation during transmission?
 - a) The strength of the device sending the signal
 - b) **The distance the signal has to travel**
 - c) The user's trust in the transmission system

2. What are some common causes of interference during signal transmission?
 - a) **Storms and neighboring wires**
 - b) Rainbows and moonlight
 - c) Animals and plants

3. What is the purpose of a repeater in signal transmission?
 - a) To create a copy of the signal
 - b) **To capture, amplify, and retransmit the same signal**
 - c) To protect the signal from interference

4. What is the purpose of an acknowledgement in signal transmission?
 - a) To inform the sender that the message was sent
 - b) **To inform the sender that the message was received**
 - c) To inform the sender that the message was encrypted

5. What is encryption used for in signal transmission?
 - a) To increase the speed of transmission
 - b) To reduce errors in transmission
 - c) **To protect the data from hackers**

Activity 2.6

Answer the questions below.

1. Give an example of a PAN you use in your daily life.

Answers will vary. Example: My smartphone connected to my Bluetooth headphones and smartwatch.

2. Give an example of a LAN you use in your daily life.

Answers will vary. Example: My computer science classroom has a switch that connects the desktops to each other and to the school server allowing them to share resources.

3. How do routers function in a network?

Routers act as gateways to the internet and control its traffic. When information has to pass outside a LAN, the router identifies its destination and the best path to get there.

Activity 2.7

Answer the questions below.

1. How is the function of a Fiber optic NIC different from an Ethernet NIC?

A Fiber optic NIC converts the electrical signals used by the device into optical signals that can be transmitted over the fiber optic cable, while an Ethernet NIC converts between the electronic digital signals of the computer and another electronic digital signals used to transfer information through long cables.

2. Why are Fiber optic NICs commonly used in data centers?

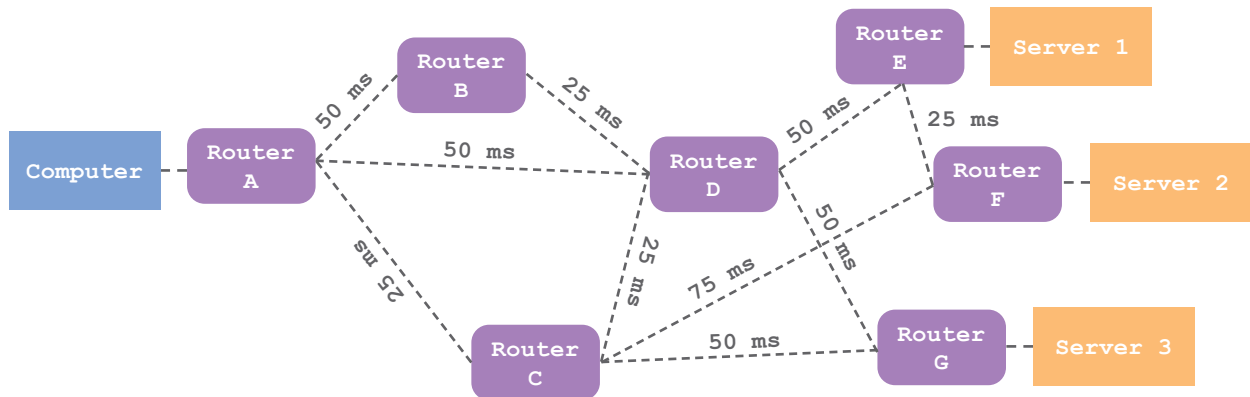
Fiber optic NICs are commonly used in data centers because they offer high-speed data transmission and reliability, which are critical in data center environments

3. Examine your personal computer and a desktop computer at school. List the various network adapters these computers have.

The answers will vary widely depending on the computers being examined.

Activity 2.8

Use the network diagram to answer the questions below.



- Complete the routing database below between Computer and Server 1.

Route Option	Hop 1	Hop 2	Hop 3	Hop 4	Total Latency
1	B: 50	D: 25	E: 50		125 ms
2	D: 50	E: 50			100 ms
3	C: 25	D: 25	E: 50		100 ms
4	C: 25	F: 75	E: 25		125 ms
5	C: 25	G: 50	D: 50	E: 50	175 ms

- Complete the routing database below between Computer and Server 2.

Route Option	Hop 1	Hop 2	Hop 3	Hop 4	Total Latency
6	B: 50	D: 25	E: 50	F: 25	150 ms
7	D: 50	E: 50	F: 25		125 ms
8	C: 25	D: 25	E: 50	F: 25	125 ms
9	C: 25	F: 75			100 ms

3. Complete the routing database below between Computer and Server 3.

Route Option	Hop 1	Hop 2	Hop 3	Hop 4	Total Latency
10	B: 50	D: 25	G: 50		125 ms
11	D: 50	G: 50			100 ms
12	C: 25	G: 50			75 ms

4. Based only on hop counts, list the best route to each server. Use latency to resolve a tie.

Computer to Server 1: Route Option 2

Computer to Server 2: Route Option 9

Computer to Server 3: Route Option 11 and 12, based on latency 12 wins.

5. Based only on latency, list the best route to each server. Use hop count to resolve a tie.

Computer to Server 1: Route Option 2 and 3, based on hop count 2 wins.

Computer to Server 2: Route Option 9

Computer to Server 3: Route Option 12

6. Use Command Prompt or Terminal to measure ping for the web address, www.google.com, and any other of your choosing.

ping <web address>

 Activity 2.9

Answer the questions below. Use an ASCII table to identify the decimal number that represents each character. Note that capital and small characters have different ASCII values and that a “space” has a value.

1. You send a message, “Hi there”, to your friend. The checksum at the recipient’s end is 704. Did the message arrive intact without error?

The checksum, based on ASCII values, is 745.

So there is an error in the message.

2. You send a message, “I finished homework”, to your friend. The checksum at the recipient’s end is 1855. Did the message arrive intact without error?

The checksum, based on ASCII values, is 1855.

So there is no error in the message.

3. A transaction code is sent between two banks, “AA103XT47”. The checksum at the recipient’s end is 551. Did the message arrive intact without error?

The checksum, based on ASCII values, is 557.

So there is no error in the message.

Activity 2.10

The program below, in Python, is intended to accept a message from the user and to calculate the checksum. Use this program to check your answers for Activity 2.9.

```
def calculate_checksum(message):
```

```
    checksum = 0
```

```
    for char in message:
```

```
        ascii_value = ord(char)
```

```
        checksum = checksum + ascii_value
```

```
    return checksum
```

```
message = input('Enter message: ')
```

```
checksum = calculate_checksum(message)
```

```
print (f'The checksum for message is: {checksum}')
```

Defines the function that calculates the checksum by adding the ASCII values.

Defines the checksum as a variable and sets it to zero.

Starts a loop that takes a character from the message each time.

Converts character to its ASCII value.

Updates checksum by adding the new ASCII value to the current checksum value.

Repeats the loop.

Gets a string input from the user and assigns it to "message".

Calculates the checksum using the function "calculate_checksum".

Prints the checksum.

Notes

The intention of this activity is to familiarize students with coding and the concept of algorithms and iteration. The intention is not to teach Python or any other coding language. Students should, however, appreciate and understand each step in the line of code.

Activity 2.11

Use a Caesar cipher wheel to answer the questions below. You may choose to construct one in class.

1. Encrypt the message “Hi there”. The key = counterclockwise 2.

Fg rfcpc

2. Encrypt the message “Beware of hackers”. The key = clockwise 5.

Gjbfwj tk mfhpjwx

3. Encrypt the message “Beware of hackers”. The key = counterclockwise 10.

Rumqhu ev xqsauhi

Activity 2.12

The program below, in Python, is intended to accept a message from the user and to calculate the checksum. Use this program to check your answers for Activity 2.9. Note that this program only performs a clockwise shift. To perform a counterclockwise shift subtract the shift value instead of adding it.

```
def caesar_cipher_encrypt(plaintext, shift):

    Ciphertext = " "

    for char in plaintext:

        if char.isalpha():
            if char.isupper():
                encrypted_char = chr((ord(char) - 65
                    + shift) % 26 + 65)

            else:
                encrypted_char = chr((ord(char) - 97
                    + shift) % 26 + 97)

            ciphertext = ciphertext + encrypted_char

        else:
            ciphertext = ciphertext + char

    return ciphertext

plaintext = input("Enter message: ")

shift = int(input("Enter shift value: "))

encrypted_message = caesar_cipher_encrypt(
    plaintext, shift)

print("Encrypted message: ", encrypted_message)
```

Defines the function that changes original plaintext to ciphertext by a specified shift.

Defines the ciphertext as an empty string.

Starts a loop that takes a character from the message each time.

Checks if the character is alphabetic.

Checks if the character is upper case.

If the character is uppercase, it calculates the ASCII value of the encrypted character by subtracting 65 (the ASCII value of 'A') from the ASCII value of the current character. The shift value is added, and the modulo 26 operation is applied to ensure the result stays within the range of the alphabet. Finally, 65 is added back to obtain the ASCII value of the encrypted character. The chr() function converts the ASCII value back to a character.

If the character is not uppercase, the same process is applied, but with 97 (the ASCII value of 'a') as the base value.

The encrypted character is added to the ciphertext string.

If the character is not alphabetic it is added as is to the ciphertext string.

Repeats the loop.

Gets a string input from the user and assigns it to "message".

Calculates the checksum using the function "calculate_checksum".

Prints the checksum.